

From Zero to Uyuni

Building a Reproducible Systems Management Lab
with openSUSE, RKE2 and Helm

*Uyuni: free, open-source software that manages many Linux
machines from one place.*

Convention Hall, UIN Sunan Kalijaga · Room 4 · 15:30

1 Phone → laptop: USB cable or 5 GHz
hotspot

2 killercode.com,
logged in

3 Don't press Start
yet

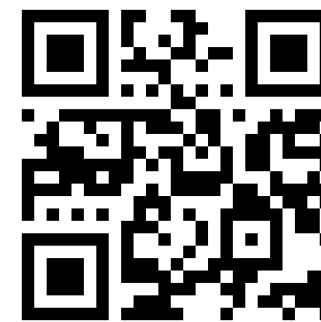


github.com/AkashKumar7902/zero-to-uyuni

Slides (spoiler-free) are in the repo

Check in now

`geeko-hq.pages.dev`



Laptop: type it, own window

Phone: scan

Nickname only: the big screen may be
photographed.

03:12

PAGER

```
$ kubectl -n uyuni get pvc var-pgsql  
Error from server (NotFound):  
persistentvolumeclaims "var-pgsql" not  
found
```

What happened to the database?

real output, captured on our server



ABOUT ME

Akash Kumar

- Software Engineer
- GSoC 2026 & 2025 · openSUSE
- LFX Mentorship 2024 · CNCF
- ICPC Regionalist

Welcome, new SRE crew

Uyuni (oo-yoo-NEE): the world's
biggest salt flat.

It runs on Salt.

HQ = a real Uyuni server.

Geeko Corp is made up. Uyuni and every error are real.



Photo: Loïc Mermilliod / CC0 / Wikimedia Commons

Your shift

1 BLUEPRINT

2 JOIN

3 ORDERS

4 OUTAGE

 FINAL
CASE

Today you are the fleet.

HOW WE PLAY



fingers at your chest



sticky on the lid



= look up

Choose your role, then START

A



field investigator

B



terminal crew

C



night school

TYPE IT ON YOUR LAPTOP

<https://killercoda.com/akashkumar7902/scenario/zero-to-uyuni>

Log in · press Start on the laptop only · never reload the tab

One crew name per pair: it belongs to both of you.



Rule 1: only inside the sandbox. A Salt master has root on its minions. Never connect your own laptop to a server you don't control.

root = Linux's all-powerful admin. Your sandbox: deleted tonight.

Nickname only · the screen may be photographed · openSUSE Code of Conduct applies ·

en.opensuse.org/Code_of_Conduct

22:00 • Tonight's first job

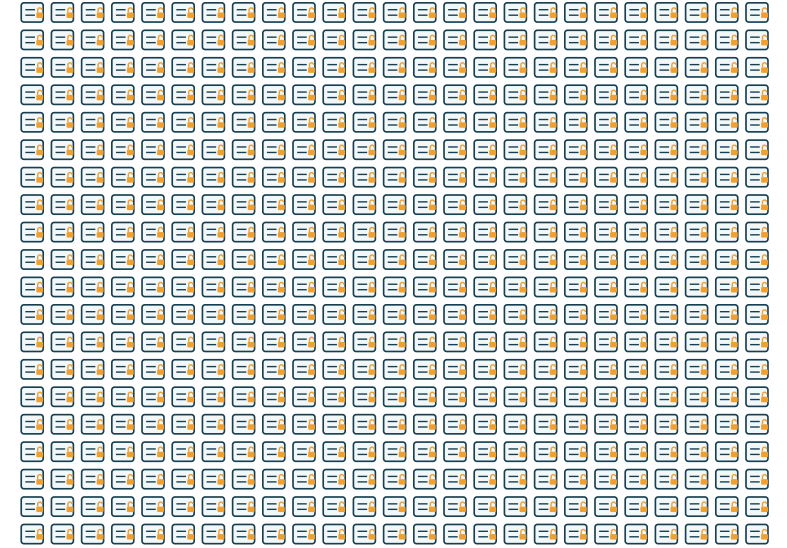
New security hole • 500 Linux machines • 1 engineer

1 log in to each

2 a script for all

3 ask 10 friends

4 go home and hope



HQ = Geeko Corp's Uyuni server

Uyuni: free, open-source software that manages many Linux machines from one place.



SEE

every machine



PATCH

install fixes



CONFIGURE

same rules
everywhere



COMMAND

one order, many
machines

free · open source · part of openSUSE · uyuni-project.org

HQ = Geeko Corp's Uyuni server

Uyuni: free, open-source software that manages many Linux machines from one place.

 SEE

 PATCH

 CONFIGURE

 COMMAND

free · open source · part of openSUSE · uyuni-project.org

Search page

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All

Physical Systems

Virtual Systems

Unprovisioned Systems

Out of Date

Requiring Reboot

Non Compliant

Ungrouped

Inactive

Systems

+ Add system

System |

Download CSV

☐	System	Updates	Patches	Packages	Extra Packages	Config Diffs	Last Che
☐	leap-a	0	0	167	0		Sep 30, 2 3:30:53 P

Items 1 - 1 of 1

Search page

Home

Systems

Salt

Images

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Software

Content Lifecycle

Audit

CVE Audit

Subscription Matching

OpenSCAP

CVE Audit ?

CVE

2026

▼

-

- ☒ Affected, patch is unavailable
- ☒ Affected, patch is unavailable in relevant channels
- ☒ Affected, partial patch available in assigned channel
- ☒ Affected, patches available in channels which are not assigned
- ☒ Affected, patches available in a Product Migration target
- ☒ Affected, at least one patch available in an assigned channel
- ☒ Not affected
- ☒ Patched
- ☒ Unknown, CVE metadata not available

Audit Servers

Audit Images

Download CSV

Filter by name

- Search page
- Home
 - Systems
 - Salt
 - Keys
 - Remote Commands
 - Formula Catalog
 - Images
 - Patches

Remote Commands

Check which minions match the target expression

Target systems (1)

leap-a

COMMAND

Salt > Remote Commands

Our real HQ

photo from Wed 30 Sep

Who runs it?

1 a normal user

2 root

3 nobody



UYUNI (oo-yoo-NEE)

Class	systems manager
Used by	teams with many Linux machines
Born	2018, from Spacewalk
SUSE's product	SUSE Multi-Linux Manager (was SUSE Manager), built from Uyuni
Home	openSUSE Project, since June 2026
Speaks	Salt
Licence	free and open source

Reference card: the only slide besides the crew words that breaks the 12-word rule.



CREW WORDS (L1-L4 = the level where you meet it)

- Uyuni manages many Linux machines from one place
- HQ our Uyuni server
- patch/CVE a fix / the ID number of a known security hole
- L1 Helm chart HQ's install recipe (the blueprint)
- L2 minion a machine HQ manages (a client): your sandbox
- L2 bootstrap . the first setup that makes it a minion
- L2 Salt how HQ gives orders; HQ = the Salt master
- L2 activation key a class code: which team you join
- L2 Salt key .. the machine's ID card; fingerprint checks it
- L3 group a named team of machines (system group)
- L3 channel ... a shelf to subscribe to: software or rules
- L3 state how a machine should look; highstate = now
- L3 remote command one order, as root, on many machines
- L4 dark not answering HQ: config, DNS, port, log

On the back of your crew card, and on your sandbox's welcome page.

Who calls first?

1 your sandbox calls HQ

2 HQ calls your sandbox

4505 • 4506

YOUR SANDBOX

a Salt minion

HQ'S FRONT DOOR

doors 4505 • 4506

HQ'S BRAIN

Salt master + database

Who calls first?

Answer: live in Room 4

INSTALL: HELM CHART = HQ'S INSTALL RECIPE

LEVEL 1 • Read HQ's blueprint

6:00 ends 15:48

TYPE IT

```
$ /root/osas26/crew.sh <your crew code>
$ helm template demo oci://registry.opensuse.org/uyuni/server-helm \
  --version 2026.8.0 --set global.fqdn=uyuni.example.test \
  > ~/uyuni.yaml
```

Q1 how many volumes?

Q2 Service salt : what type?

Q3 which images?

Evidence? compare with the manual · `/root/osas26/claim.sh <name>`

No rush: evidence stays open until 16:25. Click START at the bottom.

Blueprint debrief

Q1 volumes? _____

Q2 Salt's door type? _____

Q3 images? _____

How do you get through a locked door?

The 8 lines that open the door

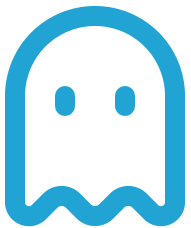
UYUNI-TRAEFIK.YAML

```
apiVersion: helm.cattle.io/v1
kind: HelmChartConfig
metadata: {name: rke2-traefik, namespace: kube-system}
spec:
  valuesContent: |-
    ports:
      salt-publish: {port: 4505, expose: {default: true}, exposedPort:
4505, protocol: TCP, hostPort: 4505, containerPort: 4505}
      salt-request: {port: 4506, expose: {default: true}, exposedPort:
4506, protocol: TCP, hostPort: 4506, containerPort: 4506}
```

Traefik = HQ's receptionist.

Speed-run: build HQ from zero

HQ from zero: **under 8 min** · one command · a free GitHub runner



Clap (or shout TRAP!) when one flies by

RKE2 v1.36.4 Uyuni 2026.08 (chart 2026.8.0) Traefik (RKE2 default) cert-manager v1.21.2
trust-manager v0.24.0 local-path v0.0.36 Helm v4.2.4
lab-only: superPrivileged



[github.com/AkashKumar7902/
zero-to-uyuni](https://github.com/AkashKumar7902/zero-to-uyuni)

HQ JOB UNLOCKED: SEE

LEVEL 2

- SEE
- PATCH
- CONFIGURE
- COMMAND

HQ JOB UNLOCKED: SEE

LEVEL 2

 SEE

 PATCH

 CONFIGURE

 COMMAND



JOIN

minion = a machine HQ manages · bootstrap = its first setup

HQ JOB UNLOCKED: SEE

LEVEL 2

 **SEE**

 **PATCH**

 **CONFIGURE**

 **COMMAND**



JOIN

minion = a machine HQ manages · bootstrap = its first setup



TICKET

activation key = a class code

HQ JOB UNLOCKED: SEE

LEVEL 2

 **SEE**

 **PATCH**

 **CONFIGURE**

 **COMMAND**



JOIN

minion = a machine HQ manages · bootstrap = its first setup



TICKET

activation key = a class code



TRUST

Salt key = the machine's ID card

HQ JOB UNLOCKED: SEE

LEVEL 2

 **SEE**

 **PATCH**

 **CONFIGURE**

 **COMMAND**



JOIN

minion = a machine HQ manages · bootstrap = its first setup



TICKET

activation key = a class code



TRUST

Salt key = the machine's ID card

Which key says your TEAM?

1 activation key

2 Salt key

I do first · doors open 15:55

WATCH

```
$ /root/osas26/join.sh --hq akash
```

Read the 4 steps.

Your banner says **YOUR OWN HQ?** Today, you are HQ: salt-key -a

15:55

LEVEL 2 • Join the fleet

5:00 ends 16:00

```
$ /root/osas26/join.sh <crew name>
```

JOIN

```
$ salt-key -L
```

LIST

```
$ salt-key -f "$(cat /etc/osas26-id)"; venv-salt-call --local  
key.finger
```

FINGERPRINT

```
$ salt-key -a "$(cat /etc/osas26-id)"
```

SAY YES

```
$ salt "$(cat /etc/osas26-id)" test.ping
```

PING

One crew name per pair. Read before you run. True? CHECK.

Front table: /root/osas26/join.sh --hq

Find your line

```
$ cat /etc/venv-salt-minion/minion.d/osas26.conf
master:
id:
grains:
  susemanager:
    activation_key:
```

Which line is HQ's address?

1

master

2

id

3

activation key

Phone < 30 %? USB-tether it: it charges.

Case 1 · The volume that never bound

- 1 waiting for first consumer to be created before binding
- 2 no persistent volumes available for this claim and no storage class is set

Which is really broken?

1 line 1

2 line 2

HQ JOB UNLOCKED: CONFIGURE · COMMAND

LEVEL 3

 SEE

 PATCH

 CONFIGURE

 COMMAND

S22.1

openSUSE.Asia Summit 2026 · Room 4

✓ BLUEPRINT

✓ JOIN

ORDERS

OUTAGE

FINAL CASE

LEVEL 3 • Orders from HQ

TYPE IT

```
$ salt "$(cat /etc/osas26-id)" cmd.run '/root/osas26/hq-say.sh welcome'  
$ salt "$(cat /etc/osas26-id)" state.apply manager_org_1.osas26-welcome  
$ cat /etc/motd
```

Uyuni's buttons = the commands you type

prompt looks odd? Press Enter.

LEVEL 3 • Orders from HQ

TYPE IT

```
$ salt "$(cat /etc/osas26-id)" cmd.run '/root/osas26/hq-say.sh welcome'  
$ salt "$(cat /etc/osas26-id)" state.apply manager_org_1.osas26-welcome  
$ cat /etc/motd
```

Uyuni's buttons = the commands you type

Still true tomorrow?

1

the command

2

the state

Quick check

No open ports. How did HQ reach you?

1 HQ called in

2 your sandbox called out, on 4505/4506

3 magic

One-breath challenge

"Uyuni is..."

- 1 a Linux distribution
- 2 software that manages many Linux machines from one place
- 3 a tool that installs Kubernetes



SEE



PATCH



CONFIGURE



COMMAND

Previously on Geeko Corp...

Uplink on



type the URL



Start



catch up

<https://killercoda.com/akashkumar7902/scenario/zero-to-uyuni>

```
$ /root/osas26/catchup.sh 1
```

```
$ /root/osas26/catchup.sh 2
```

LEVEL 4 • Incident drill

7:00 ends 16:23

WHEN I SAY GO

```
$ /root/osas26/break.sh
```

1 CONFIG

```
$ venv-salt-call --local config.get master
```

2 DNS

```
$ getent hosts <master>
```

3 PORT

```
$ timeout 3 bash -c "</dev/tcp/${cat /etc/osas26-fqdn}/4506" && echo OPEN  
|| echo BLOCKED
```

4 LOG

```
$ grep -E 'ERROR|WARN' /var/log/venv-salt-minion.log | tail -n 5
```

5 RESTART

```
$ systemctl restart venv-salt-minion
```

fix.sh when I say so. Room wins at $\leq 10\%$ dark.

Promotions • save point

Still dark? `/root/osas26/fix.sh` now

```
$ salt "$(cat /etc/osas26-id)" state.apply manager_org_1.osas26-welcome  
$ cat /etc/motd
```

**BLUEPRINT
READ**

**FLEET
JOINED**

**ORDERS
DONE**

**OUTAGE
FIXED**

Photo time!

16:27: sandbox work ends